Charger (BQ24250): Working Document

Digital Audio Player • Rev 12

Author: Alexander Loker. Design decisions, component selection, analysis and verification.

Drafted by: Claude (Anthropic). Structure and prose from the author's direction and source data.

Contents

1. Summary 1

2. Design Requirements 1

3. Pin Configuration 2

4. External Components 2

5. Equations and Results 3

6. Important Values 4

7. Open Problems 4

8. Revision Log 5

9. Sources 5

1. Summary

Field	Value
Manufacturer / MPN	Texas Instruments, BQ24250RGET (AL-PN #1-054), U202
Package	24-VQFN, 4 × 4 mm
Function	3 MHz switch-mode single-cell Li-ion charger with PowerPath (SYS output)

2. Design Requirements

Requirement	Value
Input	USB-C, 5 V
Battery	1S Li-Po, 3.0 to 4.2 V, 1850 mAh
Charge current	500 mA (0.27C)
Input current limit	1 A
V_IN_DPM	4.488 V
Switching frequency	3 MHz (2.7 to 3.3 MHz)
Charge temperature window	TS network, ≈ 2 to 57 °C (Section 5)
System peak (S4)	403 mA at 3.7 V (Power Budget)

Requirement	Value
Constraints	Low noise · Small footprint

3. Pin Configuration

Pin	Name	Connection
1	CE	CE_BQ to GPIO46; 10 kΩ pull-down (charging enabled by default)
2	EN1	10 kΩ pull-up to +3V3_DIG (R209)
3	EN2	10 kΩ pull-down to GND (R216)
4	AGND	GND
5	SDA	10 kΩ pull-up to +3V3_DIG (R211)
6	SCL	10 kΩ pull-up to +3V3_DIG (R208)
7	STAT	Green LED (D200) and 1 kΩ (R206) from SYS
8	INT	10 kΩ pull-up to +3V3_DIG (R210), INT_BQ to GPIO37
9	TS	5 kΩ from LDO; 9.88 kΩ and NTC to GND
10	ISET	R_ISET to GND
11, 12	BAT	Battery +, C_BAT
13, 14	SYS	System rail, C_SYS bank
15, 16	PGND	GND
17, 18	SW	Inductor, C_BOOT
19	IN	VUSB (filtered), C_IN, V_DPM divider
20	PMID	C_PMID to GND
21	BOOT	C_BOOT to SW
22	ILIM	R_ILIM to GND
23	VDPM	Divider tap
24	LDO	C_LDO, top of TS network
25	Thermal pad	GND

Source: [1], [3]

4. External Components

Ref	Value / Rating	MPN	AL-PN
L (L203)	1.0 µH ±20 %, 2 A I_sat, 96 mΩ, 0505	TDK TFM141206BLE-1R0MTCA	#1-063
C_PMID (C200)	1 µF 25 V X7R 0603	Murata GCM188R71E105KA64D	#1-132
C_BAT (C215)	1 µF 25 V X7R 0603	Murata GCM188R71E105KA64D	#1-132
C_LDO (C214)	1 µF 35 V X5R 0402	Murata GRM155R6YA105KE11D	#1-092
C_BOOT (C207)	33 nF 16 V X7R 0402	Murata GRM155R71C333KA01D	#1-059
C_IN (C202)	2.2 µF 25 V X7R 0603	Murata GRM188Z71E225KE43D	#1-070
C_SYS1, C_SYS2 (C208, C209)	22 µF 25 V X7S 1210	Murata GCM32EC71E226KE36L	#1-073

Ref	Value / Rating	MPN	AL-PN
C_SYS3 (C210)	0.1 µF 50 V X7R 0402	Murata GCM155R71H104KE02D	#1-080
R_ISET (R218)	500 Ω 0.1 % 0603	YAGEO RT0603BRC07500RL	#1-064
R_ILIM (R217)	270 Ω 1 % 0603	YAGEO RC0603FR-07270RL	#1-065
R_DPM1 (R202)	274 kΩ 0.5 % 0603	YAGEO RT0603DRE07274KL	#1-067
R_DPM2 (R205)	100 kΩ 0.1 % 0402	YAGEO RT0402BRD07100KL	#1-131
R2, TS (R214)	5 kΩ 1 % 0402	YAGEO RC0402FR-075KL	#1-060
R3, TS (R215)	9.88 kΩ 0.5 % 0402	YAGEO RT0402DRD079K88L	#1-061
Pull-ups / pull-downs (R208 to R211, R216, CE)	10 kΩ 1 % 0402	YAGEO AC0402FR-0710KL	#1-134
STAT LED (D200), R206	Green 0603; 1 kΩ 0.1 % 0402	Lite-On LTST-C190GKT; YAGEO RT0402BRD071KL	#1-009, #1-043
NTC	10 kΩ, β 4000 K, on cell body	n/a	n/a
F1 (FB200)	120 Ω at 100 MHz, 3 A, 0805	Würth 742792023	#1-068
C1, C2 filter (C219, C221)	4.7 µF 16 V X7R 0805	Murata GRM21BR71C475KE51L	#1-071
R1 filter damping (R227)	0.5 Ω 1 % 1/4 W 0805, pulse-rated	YAGEO PE0805FRF7W0R5L	#1-097
D1 (D201)	SMAJ7.0A TVS, SMA	Diodes Inc. SMAJ7.0A-13-F	#1-049
D2, D3 (D202, D203)	2-ch ESD, 1.5 pF (D+/D−; CC1/CC2)	TI TPD2E2U06QDCKRQ1	#1-099
J1 (J201)	USB-C receptacle, USB 2.0	GCT USB4085-GF-A	#1-069
J (J200)	Battery connector, 3-pos 1.25 mm, top entry SMD, latching (GH); 1 A per contact with 26 AWG	JST BM03B-GHS-TBT	#1-136
R_CC1, R_CC2 (R234, R235)	5.1 kΩ 0402	YAGEO RT0402BRE075K1L	#1-024

#1-009, #1-024, #1-040, #1-043 and #1-049 are library parts from other projects; tag them DAP.
Source: [3], [4]

5. Equations and Results

$$\Delta I_L = \frac{V_{BAT}(V_{IN} - V_{BAT})}{V_{IN} f_s L} = \frac{3.0 \times 2.0}{5 \times 2.7\,MHz \times 1.0\,\mu H} = 0.444\,A \quad (1)$$

Case	I_avg	I_PEAK
Charging, idle	0.500 A	0.722 A
Charging + S2 playback	0.664 A	0.886 A
Charging + S4 peak	0.903 A	1.125 A (vs 2 A I_sat)

$$R_{ISET} = 250/0.5 = 500\,\Omega;\quad R_{ILIM} = 270/1.0 = 270\,\Omega;\quad V_{IN_DPM} = 1.2 \times 374/100 = 4.488\,V \quad (2)$$

$$I_{STAT} = (V_{SYS} - 2.1\,V)/1\,k\Omega = 0.9\ to\ 2.1\,mA \quad (3)$$

Source: [1]; PE series 0805 pulse rating (≈ 200 W single pulse at 100 µs).

6. Important Values

Item	Value
Input filter insertion loss	−83 dB at 3 MHz
Filter output impedance peak	0.46 Ω with damping (46 Ω without); limit 6.25 Ω
SYS ripple (sim)	2.4 mV pp
Inductor peak, worst case	1.125 A (56 % of 2 A I_sat)
Battery FET R_ON	30 mΩ typ, 40 mΩ max
IC current limit	2.6 A min
VBUS capacitance	11.6 µF nominal; < 10 µF effective at 5 V
I2C address and bus	Shared with the DAC; 5 kΩ effective pull-ups (both sheets)
Grounding	AGND and GND on a single ground

7. Open Problems

#	Problem
1	I2C may only work while USB is present (powered from IN); verify before writing the driver.
2	NTC return shares J200 pin 3 with the battery current; attach the NTC lead at the connector end and give pin 3 a short, wide path to the ground plane near AGND.
3	Charge-temperature window (≈ 2 to 57 °C) is wider than a typical 0 to 45 °C cell charge rating; JEITA reduces charge above ≈ 44 °C. Confirm against the Jauch cell datasheet.
4	Firmware: the charger's INT and status register are the only VBUS-present indication for TinyUSB (no VBUS GPIO). Input limit is 1 A without CC sensing, above the 500 mA USB 2.0 default. Accepted (author).

8. Revision Log

Rev	Date	Change
0 to 6	2026-09-19 to 23	Initial issue through ground and VBUS checks.
7, 8	2026-09-25, 26	Reformatted; NTC open problem.
9	2026-09-27	System load from Power Budget Rev 6.
10	2026-10-03	Reviewed against the schematic: CE now a 10 kΩ pull-down (was a wire to GND). C_PMID and C_BAT #1-132, C_LDO #1-092, C_SYS3 #1-080, R_DPM2 0402 #1-131, pull-ups 0402 #1-134; STAT LED and R206 added; designators added. TS temperatures recomputed. Closed: battery connector size (author). Added: cell temperature, VBUS detection note.
11	2026-10-04	Battery connector changed to JST BM03B-GHS-TBT (GH, 1.25 mm, latching; #1-136) for the 26 AWG cell leads; pin order unchanged (BAT+, TS, GND).
12	2026-10-16	Removed the TS temperature equation and threshold table, the EN1/EN2 input-limit table, and the filter energy and pulse-power equation. I_STAT equation renumbered (3). Datasheet link added to [1].

9. Sources

1. bq2425x Product Page and Datasheet (Texas Instruments)
2. TPD2E2U06-Q1 Product Page and Datasheet (Texas Instruments)
3. DAP schematic sheet DAP-Power-2, charger and USB-C (author, 2026-10-03)
4. Alex-Loker-Altium-Database.xlsx (author)